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SMART A+ (MATHS)

CHAPTER 1

Arithmetic Sequences

Each question from 1 to 8 carries 1mark

1. Which of the numbers below can be a term of the arithmetic sequence 3,8,13,...?

A. 30 B. 31 C. 32 D. 33

2. Which of the numbers below can be the difference of two terms of the arithmetic sequence 8,12,16,...?

A. 40 B. 42 C. 45 D. 46

3. The 11^{th} term of the arithmetic sequence 18,24,30,...is ____ ?

A. 76 B. 78 C. 80 D. 82

4. The 5^{th} term of an arithmetic sequence is 27 and its common difference is 3. What is its 25^{th} term?

A. 77 B. 87 C. 91 D. 141

5. Statement 1: The 3^{rd} term of an arithmetic sequence is 11 and the 7^{th} term is 31, then its 15^{th} term 71.

Statement 2: In any arithmetic sequence ,the change in terms is proportional to the change in position.

Now choose the correct answer from those given below

A. Statement 1 is true,Statement 2 is false.
B. Statement 2 is true,Statement 1 is false.
C. Both statements are true and Statement 2 is the reason of Statement 1.
D. Both statements are true and Statement 2 is not the reason of Statement 1.

6. Statement 1: In an arithmetic sequence , the sum of two terms ,at the same distance behind and ahead a term,is twice this term.

Statement 2: If the 9^{th} term of an arithmetic sequence is 30 then, the sum of 6^{th} term and 12^{th} term is 90.

Now choose the correct answer from those given below

A. Statement 1 is true,Statement 2 is false.
B. Statement 2 is true,Statement 1 is false.
C. Both statements are true.
D. Both statements are false.

7. Statement 1: If the sum of first 7 terms of an arithmetic sequence is 210 then, its 4^{th} term is 30.

Statement 2: The sum of an odd number of consecutive terms of an arithmetic sequence is the product of the middle term and the number of terms.

Now choose the correct answer from those given below

A. Statement 1 is true,Statement 2 is false.
B. Statement 2 is true,Statement 1 is false.
C. Both statements are true and Statement 2 is the reason of Statement 1.
D. Both statements are true and Statement 2 is not the reason of Statement 1.

8 . The sum of first 8 terms of an arithmetic sequence is 320.Read the following statements about this sum

(¡) There are 4 pairs with same sum.
(¡¡) 3^{rd} term + 6^{th} term = 160
(¡¡¡) 4^{th} term + 5^{th} term = 80
(¡v) If the 7^{th} term is 50 then the common difference is 5.

Now choose the correct answer from those given below

A. (¡) and (¡v) are true
B. (¡¡) and (¡¡¡) are true
C. (¡) and (¡¡¡) are true
D.(¡¡) and (¡v) are true

3Mark Questions

9. The 5th term of an arithmetic sequence is 18 and the 8th term is 38.
   (¡) Find the 14th term of this sequence?
   (¡¡) Write the 2nd term of this sequence .

10. Consider an arithmetic sequence 7,11,15,...
   (¡) Write the common difference.
   (¡¡) Is 108 a term of this sequence? Why?

11. (¡) The 13th term of an arithmetic sequence is 82. Find the sum of its first 25 terms of this sequence?
   (¡¡) If the 6th term is 19 ,find the common difference?

12. The sum of 8th, 9th ,10th terms of an arithmetic sequence is 96.
   (¡) What is the 9th term?
   (¡¡) Find the sum of first 17 terms ?
   (¡¡¡) What is the sum of 8th term and 10th term?

13. Write the sequence of natural numbers leaving remainder 3 on division by 7. Is this an arithmetic sequence? Why?

4Mark Questions

14. The sum of 8th term and 12th term of an arithmetic sequence is 170 .
   (¡) What is the sum of 1st term and 19th term?
   (¡¡) What is the 10th term of this sequence?
   (¡¡¡) Find the sum of 19 terms of this sequence?

15. The sum of the first three terms of an arithmetic sequence is 27 and the sum of the first eleven terms is 275.
   (¡) What is the 2nd term?
   (¡¡) What is the common difference?
   (¡¡¡) Is 101 a term of this sequence? Why?

16. Consider an arithmetic sequence 5,9,13,...
   (¡) Is the difference of any two terms of this sequence be 102? Why?
   (¡¡) Find the 20th term of this sequence?
   (¡¡¡) Find the position of 241 in this sequence?

17. The 6th term of an arithmetic sequence is 57 and 9th term is 42.
   (¡) What is the 4th term ?
   (¡¡) What number should be added to the 4th term to get its first term?
   (¡¡¡) Write the first three terms of this sequence.

18. The sum of first 5 terms of an arithmetic sequence is 45 and the sum of the first 14 terms is 351.
   (¡) What is the 3rd term of this sequence?

(¡¡) What is the 10^{th} term of this sequence?
(¡¡¡) Calculate the sum of first 19 terms of this sequence?

ANSWERS

1. D (33)
2. A (40)
3. B (78)
4. B (87)
5. C
6. A
7. C
8. C
9. (¡) 14^{th} term = 38+40 = 78
   (¡¡) 2^{nd} term = 18 – 20 = -2
10. (¡) d = 4
    (¡¡) No, 108 is not a term.
    Since 108 does not leave the required remainder 3 as any term of the sequence.
    So it cannot be a term.
11. (¡) Sum of 25 terms = 25 x 82 = 2050
    (¡¡) d = 9
12. (¡) 9^{th} term = 96 /3 = 32
    (¡¡) Sum of first 17 terms = 17 x 32 = 544
    (¡¡¡) 8^{th} term + 10^{th} term = 2 x 32 = 64
13. 3,10,17,...
    Yes, d = 7
14. (¡) 1^{st} term + 19^{th} term = 170
    (¡¡) 10^{th} term = 170/2 = 85
    (¡¡¡) Sum of 19 terms 19 x 85 = 1615
15. (¡) 2^{nd} term = 27/3 = 9
    (¡¡) d = 4
    (¡¡¡) Yes, 101 leave the required remainder 1 as any term of the sequence on division by 4.
16. (¡) No, Since 102 does not leave the remainder 1 on division by 4.
    (¡¡) 20^{th} term = 5 + 19 x 4 = 81
    (¡¡¡) 60^{th} term is 241
17. d = -5
    (¡) 4^{th} term = 67
    (¡¡) 15 should be added.
    (¡¡¡) 82,77,72
18. (¡) 3^{rd} term = 9
    (¡¡) 10^{th} term = 36 (9 x 10^{th} term = 306)
    (¡¡¡) Sum of first 19 terms = 19 x 36 = 684